

CS 171 Project Proposal

Elevator Control System Safety Verification

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Topic Description

Elevator systems are safety-critical components in modern buildings, where incorrect behavior can lead to serious harm or even fatalities. Common hazards include the elevator moving while its doors are open, entering conflicting states due to improper control logic, or failing to respond to valid requests. These risks make correctness and safety essential requirements. In this project, we model a simplified elevator controller as a finite state machine, where states represent the current floor, door status, and movement direction, and transitions define the system's control logic. Using the Coq (Rocq) proof assistant, we formally specify and verify key safety and correctness properties, providing strong guarantees about the system's behavior beyond what testing or simulation alone can ensure.

Desired Properties to be Verified

- **Door Safety:** The elevator shall never move while its doors are in an open state; a transition to a moving state is only permitted when the door is confirmed closed.
- **Movement Safety:** The elevator shall not transition to a moving state if it is already at the requested destination floor, preventing unnecessary or hazardous motion.
- **State Consistency:** Every reachable state of the system must satisfy a well-formedness invariant: floor indices are within valid bounds, and motion direction is consistent with the door and movement status.
- **Determinism:** Given the same current state and input command, the transition function must produce exactly one successor state; no ambiguity or nondeterminism is permitted in the control logic.
- **Totality:** The transition function is defined for all valid combinations of states and inputs, ensuring the system never encounters an undefined or unhandled situation during operation.
- **Liveness:** Under fair scheduling assumptions, the elevator must eventually service any pending floor request; the system shall not indefinitely ignore or defer a valid call.